# TradeCompare – Project Documentation

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## 1. Project Overview

TradeCompare is a web platform designed to help U.S. investors evaluate and compare online stock trading platforms. The site presents critical data, including fees, minimum deposits, platform features, and trust ratings. Users can interact with sortable and paginated tables, visual charts, and dynamic carousels, making the experience both informative and intuitive.

The platform includes educational resources to help investors understand key trading metrics, a contact section for transparency, and responsive design with a dark mode option to ensure accessibility across devices.

## 2. Site Structure and Architecture

TradeCompare follows a frontend-backend separation, with interactive features handled by JavaScript and data served via a Node.js/Express API.

### Frontend Structure

- **HTML/CSS**:

- Semantic HTML ensures accessibility and SEO optimization.
- CSS Flexbox and Grid layouts create responsive designs for desktop, tablet, and mobile devices.
- Dark mode toggle provides a user-preference option.

- **JavaScript**:

- Dynamically fetches platform data from /api/platforms.
- Handles table sorting, searching, and pagination on the client side.
- Renders Swiper.js carousels for platform highlights.
- Generates interactive charts using Chart.js, including ratings and Forex market trends.

## Backend Structure

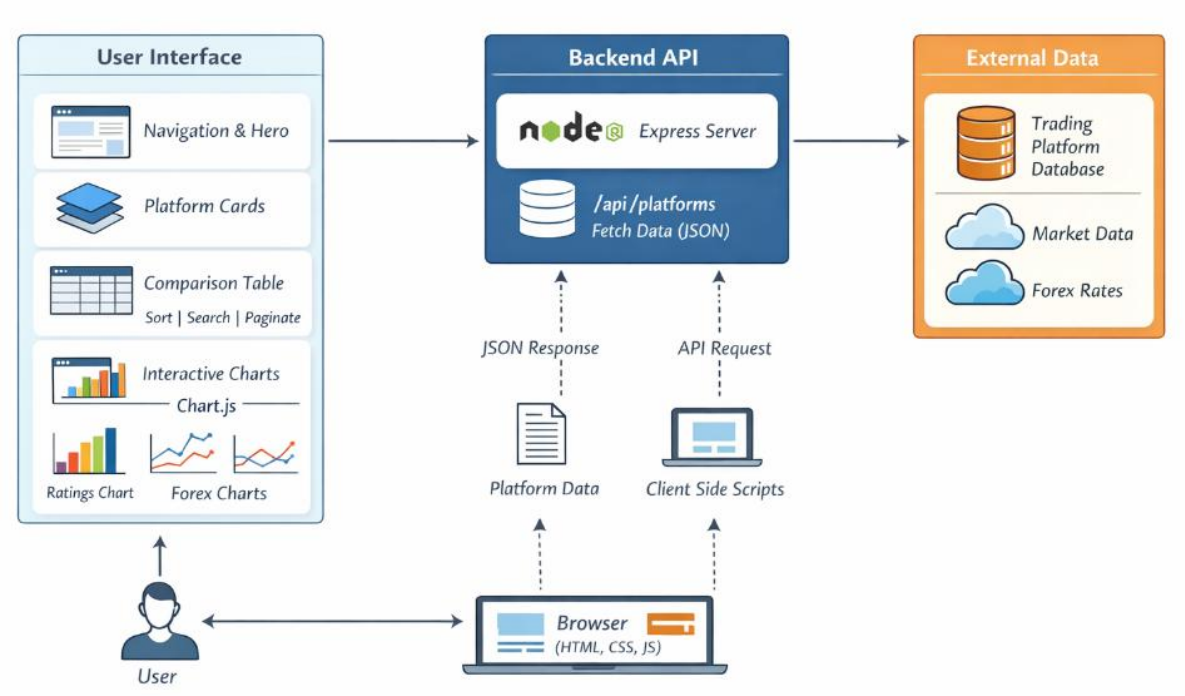
- **Node.js + Express**:

- Serves platform data in **JSON format** via /api/platforms.
- Allows easy updates to platform information without changing frontend code.

- **Data Handling**:

- JSON includes essential fields: platform name, fees, minimum deposit, best use case, rating, and URL.
- Frontend uses this data to populate both the table and the carousel dynamically.

*Diagram for Architecture Visualisation*



# 3. Design Decisions and Trust Building

TradeCompare is designed to provide trust and credibility for U.S. investors:

- **Professional Layout**:
  - Clean, structured sections including hero, comparison table, carousel, charts, education, and contact.
- **Transparency**:
  - Data rendering, clear fees, and direct links to platform websites.
- **Educational Focus**:
  - Explanatory cards help users understand fees, minimum deposits, and platform tools.
- **Interactive Features**:
  - Sortable and searchable tables, visual charts, and carousel highlight key metrics
- **Contact Section**:
  - Presents key information like email, phone, and social links

# 4. Technical Implementation Highlights

- **Comparison Table**:

  - Dynamically renders rows using JavaScript.
  - Implements client-side **pagination (5 rows per page)**, search, and sorting.
  - Sticky headers and hover effects improve readability.

- **Cards Carousel (Swiper.js)**:

  - Highlights top platforms with a responsive, looped slider.
  - Autoplay and clickable navigation enhance interaction.

- **Charts (Chart.js)**:

  - Ratings Chart: Shows trust scores for platforms.
  - Forex Charts: Visualize USD/EUR, USD/JPY, GBP/USD trends.

- **Responsive and Dark Mode**:

  - Media queries adjust layouts for mobile and tablet.
  - Dark mode changes background and text colors for night-time usability.

- **Backend JSON API**:

  - Simplifies updates without requiring backend database changes.

# 5. Maintenance Strategy

Daily and periodic site maintenance ensures consistent reliability:

- **Data Updates**:

- JSON platform data can be updated easily for fees, ratings, and minimum deposits.

- **Performance Monitoring**:

- Check table sorting, pagination, and carousel functionality.
- Verify charts render correctly across devices.

- **Backups**:

- Maintain backups of platform JSON data to prevent data loss.

# 6. Future Enhancements

- Integrate a database to support automated platform updates.
- Add user accounts for personalized tracking and recommendations.

# 7. Conclusion

TradeCompare is a trusted, user-friendly platform for evaluating U.S. stock trading services. Each component of the site, from the interactive, sortable tables to the dynamic charts that visualize ratings and market trends, the responsive carousel cards highlighting top platforms, and the educational content sections explaining key investment concepts, was carefully designed to deliver a smooth, intuitive, and engaging user experience. Every feature works together to help investors make informed decisions quickly, while maintaining transparency, accessibility, and credibility throughout the platform.